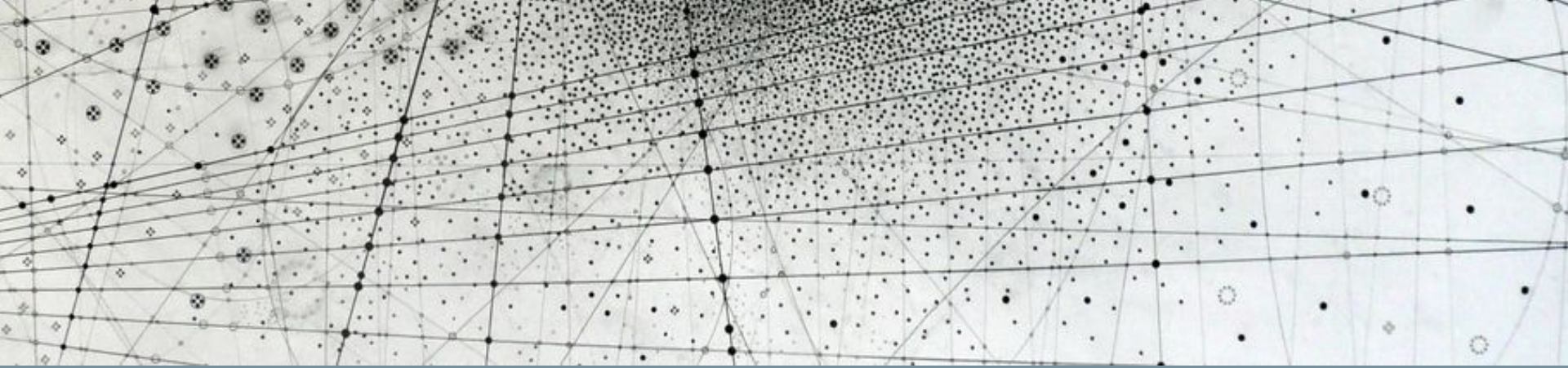
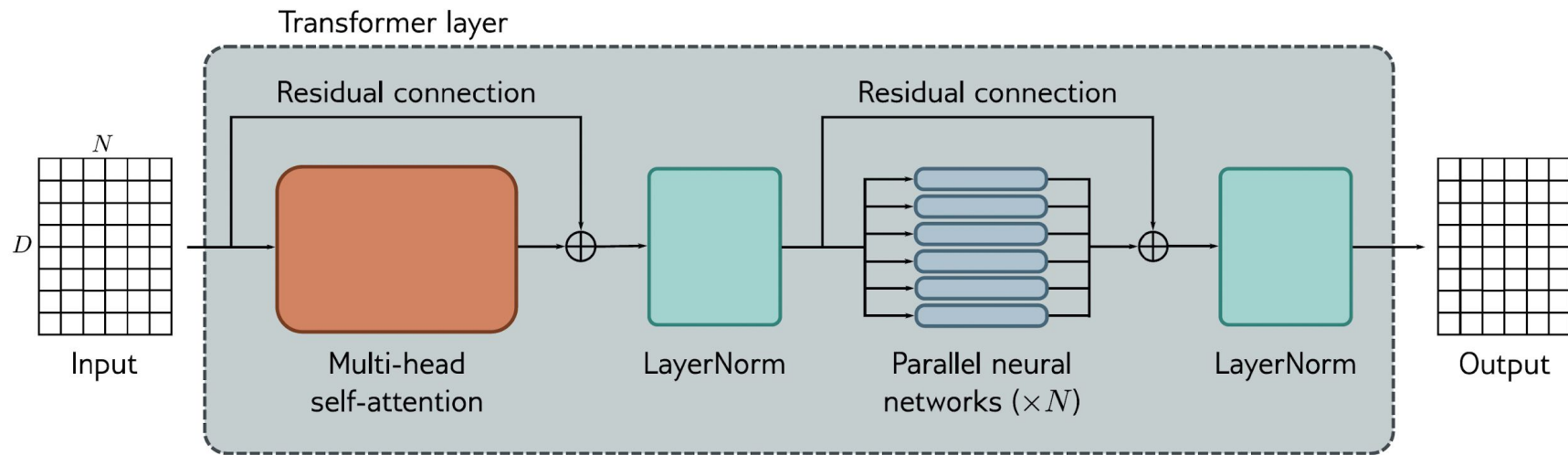




Research Methods in Machine Learning & Physics

Quick recap

Anatomy of a Transformer layer



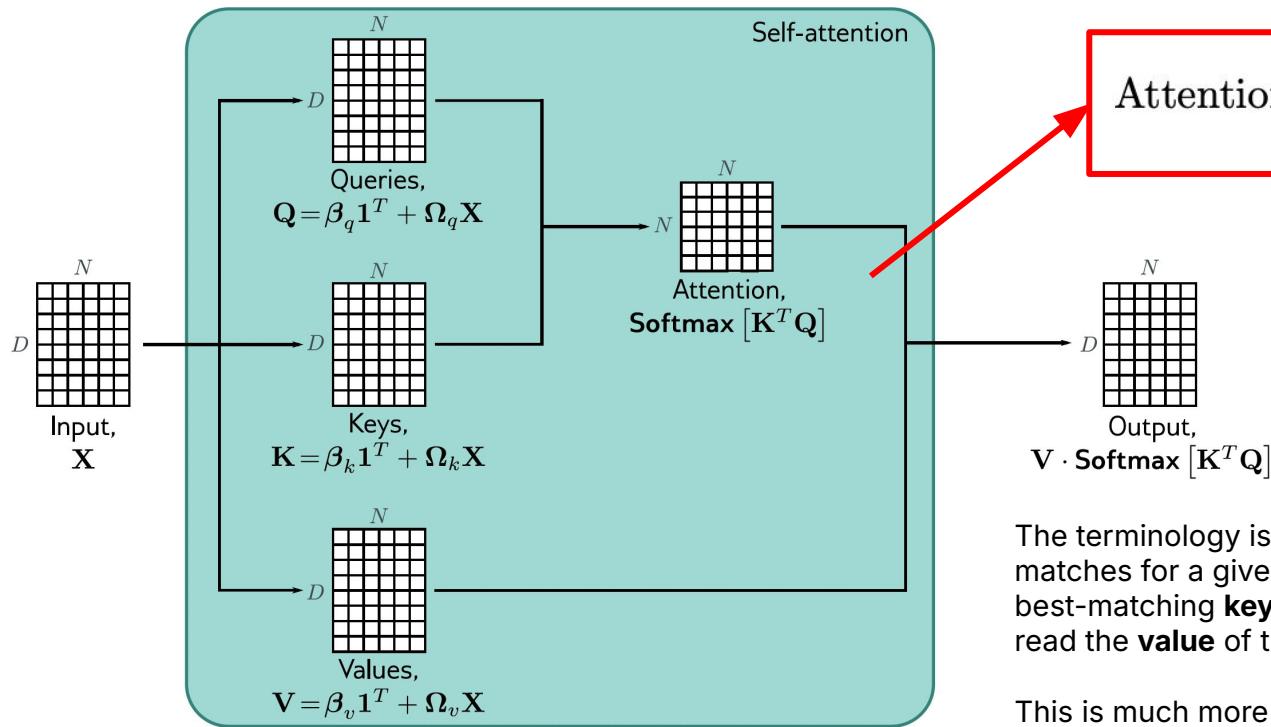
Learning a richer
representation of your
input data

Asking useful questions
from your embeddings

Inside the self-attention block...

- The goal of self-attention is to find important relationships between pairs of inputs → **dot product**
- We want to do this in a **computationally-efficient** manner
 - Easily scalable
- We want it to be able to process **variable-length inputs**
- And we also want this to reflect **permutation equivariance**

Inside the self-attention block...



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The terminology is inspired by searching a database for matches for a given query: first, you find the best-matching **keywords** for your **query**, and then you read the **value** of that corresponding file.

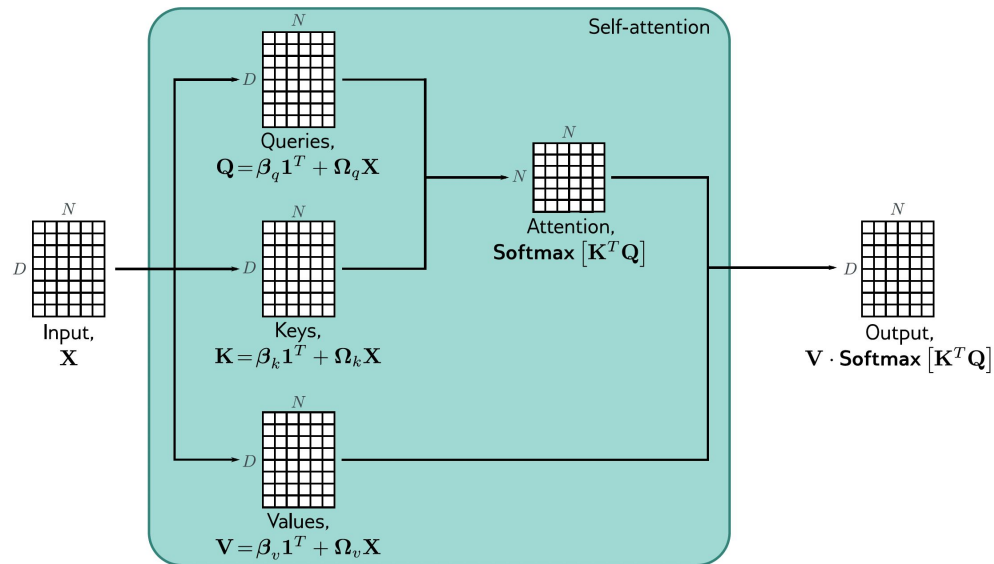
This is much more flexible & dynamic than simply taking the similarity of two inputs – it expands this to a **context-dependent framework** (based on your query Q).

Transformers, Part II

Tokenization

We saw from these diagrams that the inputs to the transformer are in a fixed size D .

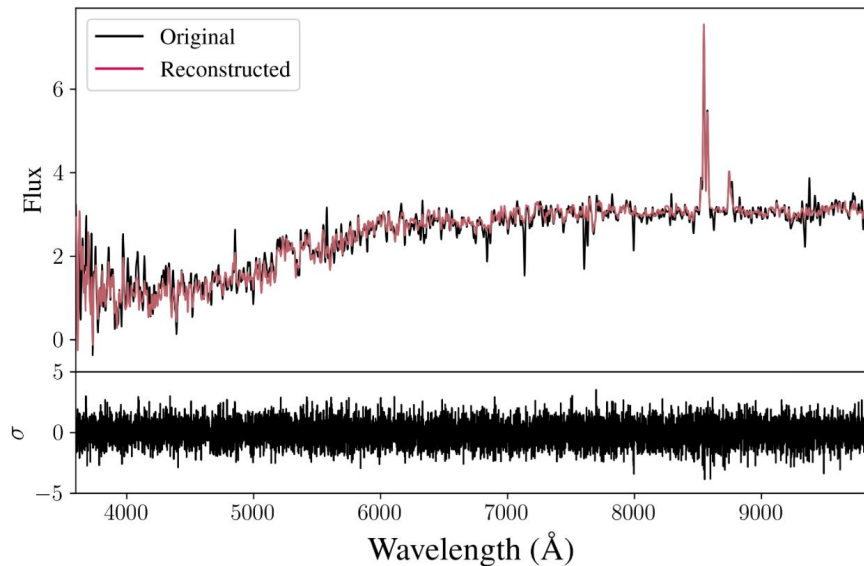
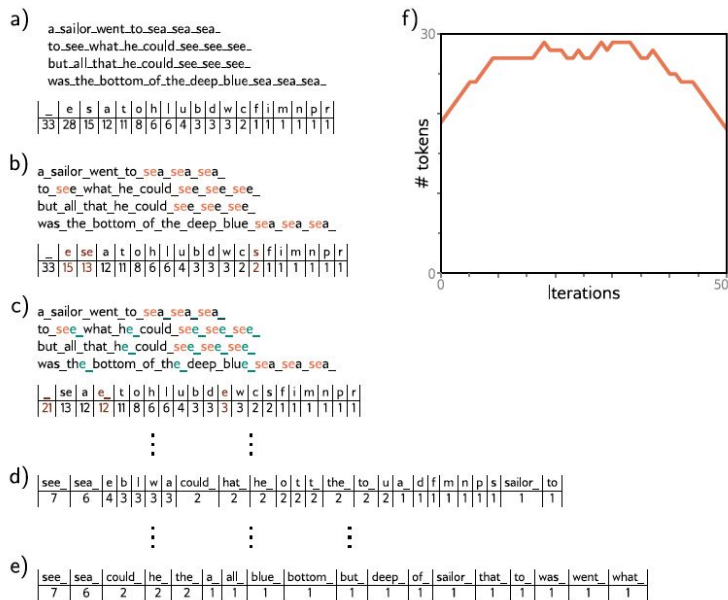
How do we convert our input data into these “tokens”?



To	date	I	the	cle	ve	rest	thinker	of	all	time	was
5.4	7.8	9.7	2.6	3.6	5.6	1.6	9.7		3.2	6.7	4.4
7.1	5.2	7.9	7.7	4.3	4.3	1.1	4.6		6.6	2.7	8.4
6.0	5.6	4.6	4.5	6.9	9.8	6.5	9.7		1.3	7.3	6.9
5.4	9.2	7.7	5.6	0.6	1.0	1.4	6.0		7.1	9.5	2.9
4.2	0.7	1.2	0.2	6.6	2.1	1.9	7.3		2.9	2.5	8.1
6.4	0.9	6.3	6.1	6.6	1.6	3.7	0.4	...	1.8	5.7	3.9
4.3	0.2	1.4	6.1	2.1	6.5	8.1	2.8		5.8	5.9	8.7
8.8	8.2	9.4	6.1	1.3	2.5	1.0	1.2		0.2	5.7	5.8
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮		⋮	⋮	⋮
3.8	8.6	4.1	6.8	3.6	2.4	1.0	1.2		0.0	9.4	6.9

Tokenization

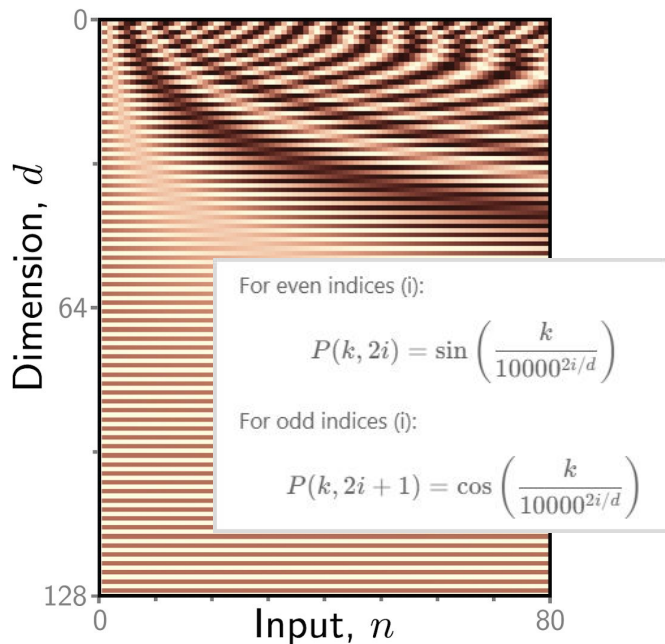
In practice, you can either use a pre-trained tokenizer for common modalities (e.g. text, images). If you have a custom data format or modality, though, you might opt to train your own tokenizer:



Positional encodings

Absolute

e.g. sinusoidal:



Relative

- RoPE = Rotary Position Embedding

ROFORMER: ENHANCED TRANSFORMER WITH ROTARY POSITION EMBEDDING

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November 9, 2023

- ALiBi = Attention with Linear Biases

TRAIN SHORT, TEST LONG: ATTENTION WITH LINEAR BIASES ENABLES INPUT LENGTH EXTRAPOLATION

Ofir Press^{1,2} Noah A. Smith^{1,3} Mike Lewis²

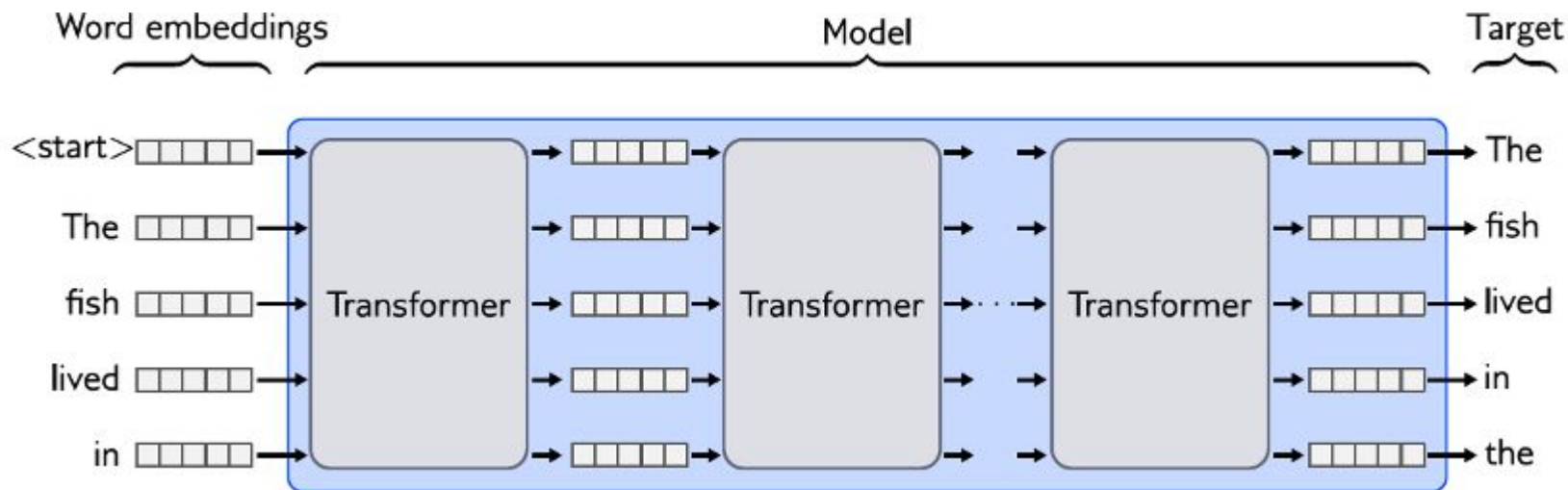
¹Paul G. Allen School of Computer Science & Engineering, University of Washington

²Facebook AI Research

³Allen Institute for AI

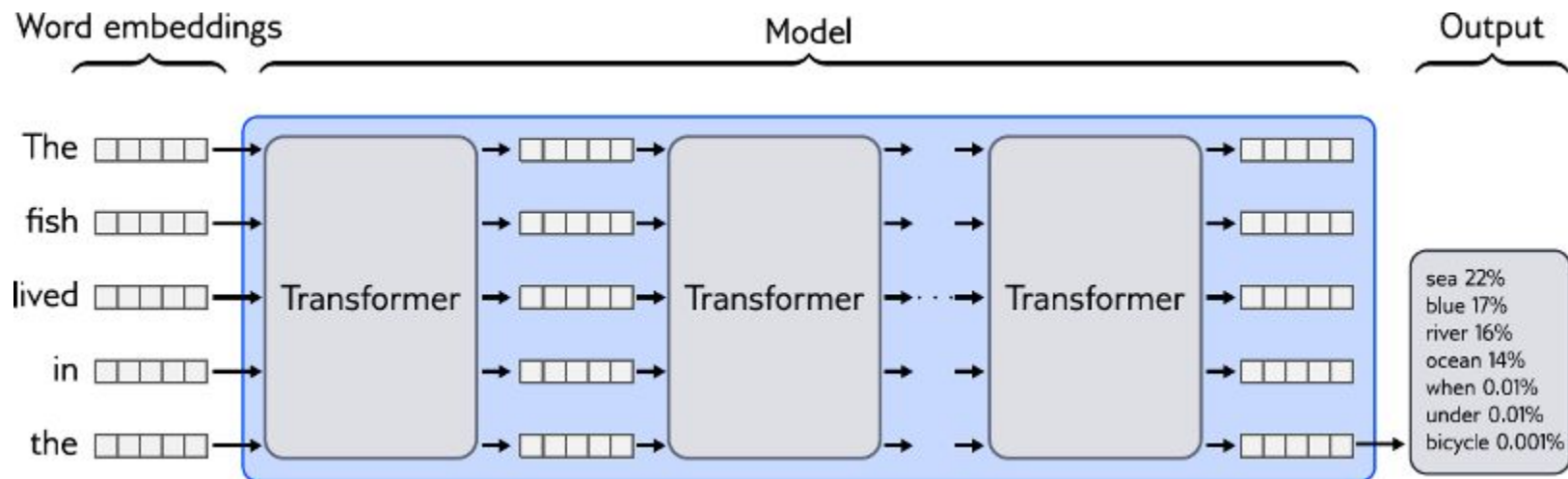
ofirp@cs.washington.edu

Predicting all at once



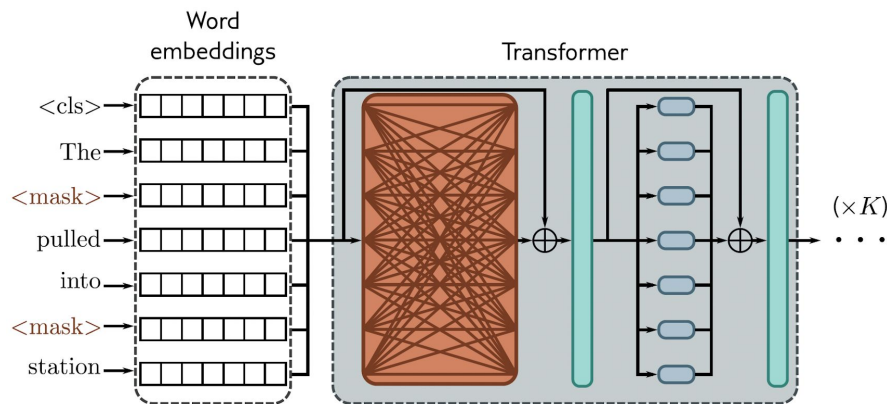
Masked Attention

Autoregressive = predict the next token

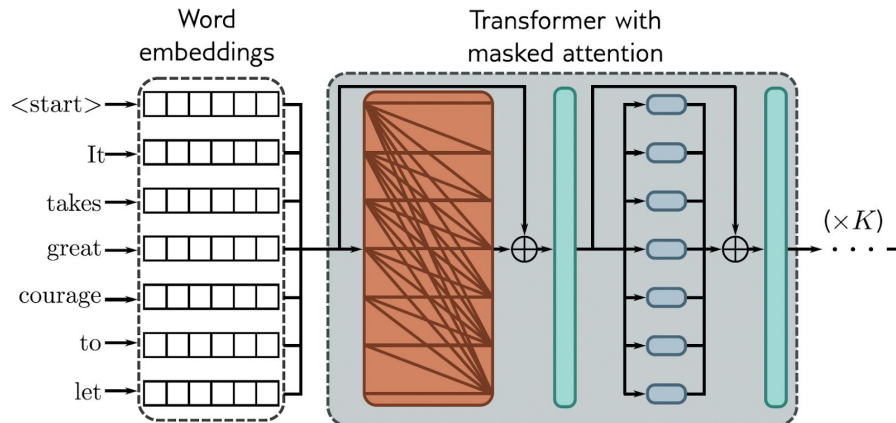


Encoder & Decoders

Encoder (e.g. BERT)



Decoder (e.g. GPT-5)



Encoder-Decoder (translation) – put these two together

In-class work

The Annotated Transformer:

<http://nlp.seas.harvard.edu/annotated-transformer/>

(Focus on *understanding* the code cells, not just *executing* them.)

Some specific tasks:

- **Explain the attention mechanism to your partner** (without looking at the code)
- Look at the **mechanics of multi-head attention**: how does the splitting & concatenation work, in practice?
- Why did the authors choose **sine & cosine positional encodings**?